

Coupling-gated cortical coherence: a conditional framework for unity and self-representation

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Abstract

Coordinated cortical activity need not describe a single experienced situation. We develop a conditional field account of coherence, compatibility and self-representation. Its demonstrated results establish what coordination leaves undetermined: increasing coupling, decreasing diffusion and a matched common drive can produce identical stationary phase distributions; equal phases permit incompatible local states; and a plasticity control preserves coherence while losing alignment with a specified environmental template. The conditional construction glues compatible spatial measures into one state and makes that state the unique fixed point of a contracting reconstruction map. Lean 4 checks the composition under eight connecting hypotheses and additional structures, with compatibility and accuracy at the glued state supplied as premises. We propose the glued state as a formal correlate of conscious unity and its accurate internal reconstruction as a correlate of a minimal self. The physical hypothesis is that changing extracellular geometry increases field-mediated coupling during awakening. A noisy mean-field phase model gives a square-root stationary onset, but finite ramps delay the response and the distributional

equivalence requires measurements that distinguish recovery mechanisms. Testing the account further requires identifying neural content and self-encoding independently, then determining whether coordination maintains their agreement and whether these relations matter for unified experience.

1 Introduction

How could activity distributed across cortex support one experience? A field account proposes that endogenous extracellular electromagnetic fields partly coordinate spatially separated neural processes (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Yet even successful coordination leaves a question: what makes the coordinated activity a description of one experienced situation, with a perspective on that situation?

We propose an account with three requirements. *Coherence* concerns whether coupling sustains collective activity. *Compatibility* concerns whether local descriptions agree where their domains overlap. *Self-representation* concerns whether the resulting global state supports an accurate model of itself. The latter two express commitments about consciousness whose motivation and possible dissociations we develop below.

Consider the experience of reaching for a cup. A description of the cup's location and a description of the hand's relation to it can share information about where the cup is while contributing different contents to the episode.

19 Synchronous activity in the two processes would leave any disagreement
20 about that shared quantity unresolved. The formal counterexample isolates
21 this problem using spatial mass profiles at a common phase that disagree
22 on an overlap. We return to the cup example to ask what agreement would
23 mean for perceptual content, and what an internal representation would have
24 to add to place the perceiver in that situation. It is an illustration of the
25 modelling task, not a neural model implemented here.

26 Awakening gives the physical requirement a concrete setting. We hypoth-
27 esize that changing extracellular geometry slowly increases effective coupling
28 through a synchronization threshold. Under stationary branch tracking, a
29 noisy phase model fixes a square-root onset of coherence. The experiment
30 faces two distinct obstacles. Finite ramp speeds delay the response, so a
31 fitted onset exponent depends on the drive. Moreover, successive stationary
32 phase distributions can be identical under increasing coupling, decreasing
33 diffusion or a suitably matched common drive. Recovery must therefore be
34 measured in a way that identifies dynamics and distinguishes causes, rather
35 than inferred from the shape of a coherence trace alone.

36 A plasticity control tests another route to representation. Its squared-
37 drift objective rewards equalizing drift, which in the chosen frequency-cluster
38 model favors connections across the clusters that define the diagnostic tem-
39 plate. Coherence survives while alignment with that template declines. To-
40 gether with the overlap counterexample, this result establishes a limit on
41 what the coordination model explains. Compatible local content needs its

own mechanism and observable.

The contribution has three levels. The separation results establish limits on what coordination explains. The conditional construction gives a unique global state from specified compatible spatial measures and makes it the unique fixed point of a contracting map assumed to reconstruct that state. Lean 4 checks the composition, and toy witnesses establish joint satisfiability of its assumptions. The biological interpretation proposes that the glued state and its internal reconstruction capture conscious unity and a minimal self. This last step requires identifying meaningful neural descriptions and explaining how coordination maintains their agreement. Figure 1 locates the proposed connections, and Section 6 states the physical and thermodynamic assumptions.

2 Coherence: the coupling model

2.1 Dynamics and the stationary threshold

The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

Here $D > 0$ is phase diffusion, W_i are independent Wiener processes, and K_{ij} has inverse-time units, with angles dimensionless. The uniform mean-field convention is $K_{ij} = K/N$. Frequency heterogeneity and phase diffusion are

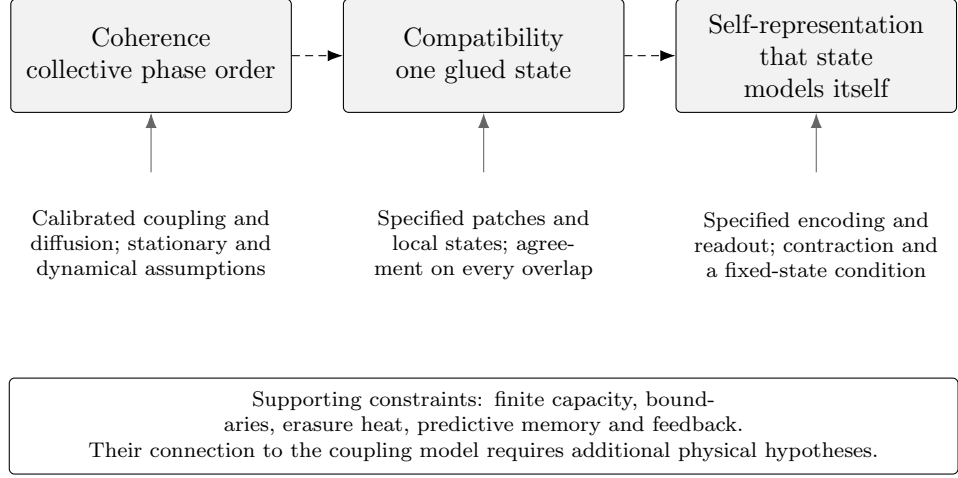


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the eight implications consumed by each branch of the formal composition.

60 separate parameters.

61 For identical frequencies in their rotating frame, the stationary ansatz is

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

62 where I_n are modified Bessel functions and ψ is the collective phase. Lean
 63 verifies normalization, the circular order parameter and self-consistency for
 64 this density. At $K \leq K_c = 2D$, the only nonnegative self-consistent order
 65 parameter is zero. At $K > K_c$, exactly one positive solution coexists with

66 zero; it emerges continuously and increases strictly with K . Near threshold,

$$\frac{r^2}{K - K_c} \rightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

67 These results characterize the stationary branch (Fig. 2). Connecting it
 68 to finite trajectories requires a dynamical mean-field limit and dynamical
 69 selection, neither proved here. Propagation of chaos concerns the first of
 70 these steps (Sznitman, 1991); exchange symmetry alone does not establish
 71 independence, especially when realizations have an unpinned collective phase.

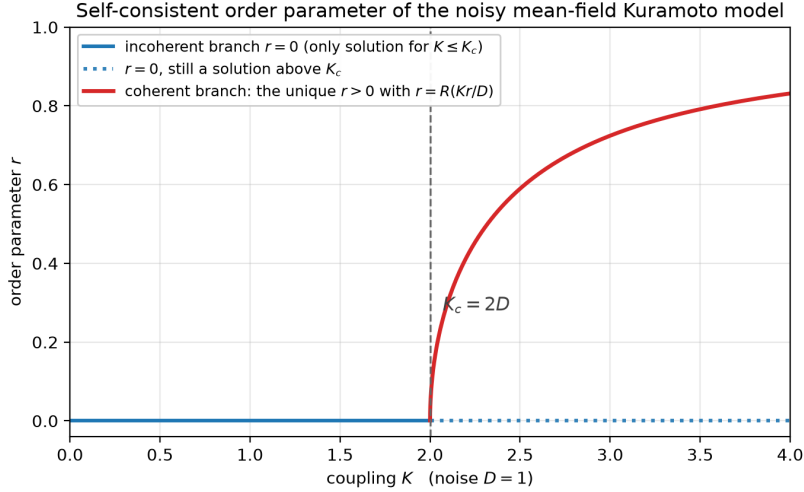


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

72 A spatial version replaces the finite sum by an integral against a kernel
 73 $K(x, y)$. The substrate measure has total mass one, making the mean-field

74 strength a kernel average. Applying the scalar threshold to a heterogeneous
 75 or signed kernel requires a justified reduction to that mean-field description.

76 **2.2 Physical identification and calibration**

77 The cortical hypothesis identifies part of the effective coupling with endoge-
 78 nous extracellular electromagnetic interactions. Fields can modulate spike
 79 timing (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field
 80 models have been proposed for coordinating distributed neural information
 81 (Pinotsis & Miller, 2022; McFadden, 2020). The quantitative question is
 82 whether that interaction supplies sufficient coupling relative to diffusion.
 83 Changes in phase organization under propofol (Bastos et al., 2021) moti-
 84 vate studying this regime, but do not measure the coupling itself.

85 The physical field predicate requires a jointly continuous kernel, a nor-
 86 malized substrate, a mean-field strength equal to the spatial average, and no
 87 single site providing more than half that strength. Regular mesh refinement
 88 establishes convergence of a scalar coupling-energy quadrature; constructing
 89 the kernel and relating it to trajectories are separate modelling tasks. The
 90 formal witnesses supply toy instances of the predicate.

91 Calibration is essential because a spike-time shift is not a coupling rate.
 92 An estimate proportional to $N\Delta tf$, using population size, timing shift and
 93 oscillation frequency, is dimensionless. A conversion of the form

$$K_{\text{eff}} = \gamma N s E f \tag{4}$$

94 requires a rate γ , where s is the timing shift per unit field and E the field
 95 amplitude. This rate specifies how induced timing shifts persist and accumu-
 96 late into phase drift. Supplemental Material explains why the other factors
 97 cannot determine it. The biological threshold test requires independent es-
 98 timates of the interaction kernel, its aggregation, and phase diffusion in the
 99 same cortical regime.

100 **3 Testing coherence through recovery after** 101 **awakening**

102 The proposed control variable is extracellular geometry. Sleep-associated
 103 changes in extracellular space and clearance (Xie et al., 2013) motivate a
 104 coupling parameter that evolves more slowly than phase dynamics. The
 105 extracellular volume fraction is approximately $\alpha = 0.2$ at diffusion tortuosity
 106 $\lambda \approx 1.6$ (Syková & Nicholson, 2008); the cited sleep study reports a rise
 107 in α from roughly 0.14 to 0.23 (Xie et al., 2013). Below 10 kHz cortical
 108 tissue conducts approximately resistively at $0.3\text{--}0.6\text{ S m}^{-1}$ (Barbour, 2017).
 109 In an effective-medium description at fixed transmembrane current density,
 110 a larger extracellular volume fraction raises conductivity and reduces field
 111 amplitude.

112 This gives the coupling-gated hypothesis a directional prediction: contrac-
 113 tion of extracellular space on awakening raises E and, with the other factors
 114 in Eq. (4) held fixed, raises effective coupling. The proposed geometry–

115 coupling relation and its role in human recovery both require measurement.
 116 The effective-medium argument supplies a sign under stated conditions; cal-
 117 ibration supplies the magnitude and the mapping to the scalar model.

118 If $K(t)$ rises through K_c and the phase distribution tracks the coherent
 119 stationary branch, Eq. (3) gives

$$r(t) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad t > t_0, \quad \dot{K}(t_0) > 0. \quad (5)$$

120 The crossing time satisfies $K(t_0) = K_c$. It is a prediction of latency only when
 121 coupling is measured independently of coherence. The subsequent shape
 122 and saturation depend on the later coupling trajectory. Critical slowing
 123 down restricts branch tracking near threshold, while finite populations have
 124 fluctuating order below it. A measurable onset therefore needs an explicit
 125 criterion above that fluctuation floor.

126 **3.1 What a recovery distribution can identify**

127 Equation (2) gives a concentration–coherence relation with no fitted shape
 128 parameter. Testing it requires separate estimators: r from circular resultant
 129 length and a from density shape. Estimating a by inverting R from the
 130 measured r makes agreement automatic. Even a valid agreement tests a
 131 statistical description shared by the von Mises family; it does not determine
 132 the coupling medium.

133 There is a stronger limitation. Successive equilibrated phase distributions

134 cannot, by themselves, distinguish three different recovery mechanisms. Let
 135 $q(t)$ be a prescribed ratio crossing 2, let $r(q)$ denote its stationary branch,
 136 and choose a fixed rate $D_0 > 0$. Increasing coupling with $K(t) = D_0 q(t)$
 137 and $D = D_0$ gives the same stationary density as fixed coupling K_0 with de-
 138 creasing diffusion $D(t) = K_0/q(t)$. Both have concentration $a = q(t)r(q(t))$.
 139 Uncoupled oscillators under a common prescribed drift $-h(t) \sin(\theta - \psi)$, with
 140 $h(t) = D_0 q(t)r(q(t))$ and $D = D_0$, have that same density. Their distribu-
 141 tions coincide window by window, even though their physical causes differ.
 142 Supplemental Material supplies the construction and its numerical feasibility
 143 check.

144 The missing observable is the physical clock. Resolved phase increments
 145 contain rate information: in the ideal continuous-path model, quadratic vari-
 146 ation accumulates at rate $2D$, while conditional increment means constrain
 147 drift. This can separate coupling and diffusion under the specified observa-
 148 tion model. Distinguishing endogenous coupling from the matched common
 149 drive additionally requires a measured drive or a controlled perturbation, be-
 150 cause their mean-field drifts can coincide. Identifying the coupling as electro-
 151 magnetic then requires evidence specific to that interaction. This hierarchy
 152 determines the experiment: distribution, dynamics and physical mechanism
 153 are distinct targets.

154 **3.2 Finite ramps change the observable onset**

155 Finite simulations characterize how the stationary prediction can appear in
156 a recovery measurement. The ramp study uses $N = 2000$, $D = 1$ and 32
157 replicas at four coupling speeds. Escape is the first of three consecutive
158 saved samples at $r \geq 0.2$ after threshold. The supplement gives parameters,
159 estimators and further controls.

160 Faster ramps delay escape (Fig. 3). Only 5/32 replicas escape within the
161 fastest window. The three slower ramps have mean coupling delays 0.1947,
162 0.0684 and 0.0253. Their fitted onset exponents run from 0.898 to 0.417,
163 compared with 0.443 to 0.448 from the same estimator on the exact stationary
164 branch at the same coupling grids. At the fastest speed the ensemble mean
165 never reaches the declared fit window; the returned value 0.964 is a local
166 log-slope of a delayed trajectory.

167 This is delayed passage through a noisy bifurcation (Berglund & Gentz,
168 2002), with a direct consequence for the awakening test: the apparent expo-
169 nent depends on the ramp and on the fitting window. A deviation from the
170 square-root law constrains the adiabatic approximation only where branch
171 tracking is independently justified. The sweep does not bracket the specified
172 deviation-based critical ramp speed.

173 Additional controls test the approximations used by the physical model.
174 Time-step refinement attributes most of the sampled discrepancy from the
175 stationary order to integration bias. Population-size diagnostics show why
176 a shared random collective orientation can preserve unconditional pair cor-

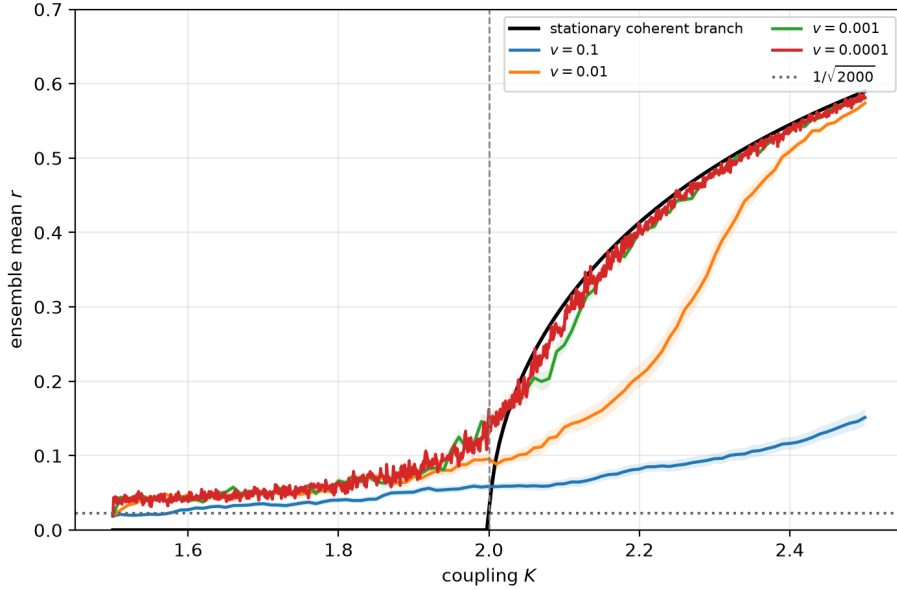


Figure 3: Finite-rate departures from the stationary branch. Ensemble-mean order versus coupling for four ramp speeds v , with one-standard-error bands across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape, changing the apparent recovery onset.

177 relation in the coherent regime. Spatial and signed-network controls exhibit
 178 coherence at specified total coupling, with the refined signed-network cross-
 179 ing bracket $K_{\text{eff}}/D \in [1.958, 2.009]$ still containing 2. These controls delimit
 180 finite regimes and numerical resolution; the supplement reports their full
 181 evidence. Calibrating the cortical coupling remains the experimental task.

182 3.3 What the estimators produce on EEG

183 Public propofol-sedation EEG (Bajwa et al., 2025) provides an exploratory
 184 exercise of the concentration and coherence estimators. The implemented

185 analysis pools channels over 100-ms windows, estimating a spatiotemporal
186 phase distribution. Bipolar and common-reference montages produce differ-
187 ent residuals. The bipolar results are compatible with the Bessel relation,
188 but the concentration range is too narrow to distinguish it from its linear
189 approximation at the observed scatter.

190 An informative recording needs a broader concentration range, estima-
191 tor calibration that accounts for dependence, and an observation model for
192 source mixing, referencing and temporal pooling. The sleep-recovery hypoth-
193 esis remains untested by these sedation recordings.

194 **3.4 A protocol for the awakening experiment**

195 The analysis defines six requirements for the physical test.

- 196 **1. Coupling and diffusion in the same units.** Estimate K and D
197 from resolved phase dynamics in the same cortical regime, separating
198 diffusion, frequency heterogeneity and observation noise. A measured
199 drive or controlled perturbation is additionally needed to distinguish
200 endogenous coupling from the matched common-drive alternative.
- 201 **2. Geometry calibrated to coupling.** Establish the relation between
202 an extracellular-geometry observable and effective coupling, with un-
203 certainty. A diffusion-MRI or astrocytic proxy for volume fraction re-
204 quires its own validation; the conductivity argument supplies only the
205 expected sign.

206 **3. Adequate phase measurements.** For the proposed spatial estima-
207 tor, use at least 100 sites per estimate, r from resultant length, and
208 a from unweighted regression of log density on the cosine of centered
209 phase. This numerical minimum needs null calibration at the actual
210 sample count, concentration range and dependence. Temporal pooling
211 changes the estimand.

212 **4. An independent recovery endpoint.** Measure behavioral recovery
213 separately from the phase trace.

214 **5. A predeclared onset criterion.** Fix a coherence threshold above the
215 finite-size floor and account for finite ramp speed before interpreting
216 an onset exponent.

217 **6. Matched model comparison.** Compare coupling-gated recovery
218 with diffusion change, common drive and alternative slow-recovery
219 models under the same observation and filtering assumptions. Station-
220 ary distributions alone cannot resolve the equivalence in Section 3.1.

221 Failure of the concentration–coherence relation over an informative range
222 would constrain the phase-density model. Failure of the calibrated geometry–
223 coupling association would constrain the proposed physical gate. A com-
224 peting dynamical explanation that better predicts independently measured
225 recovery would count against its role in awakening. These tests address the
226 first requirement of the framework. Even a favorable outcome leaves the
227 question of what the coordinated activity represents.

228 4 Compatibility: from coordinated activity 229 to a global description

230 4.1 Why common timing is insufficient

231 Consider overlapping patches U and V , each carrying a local description.
232 Compatibility requires that restricting either description to $U \cap V$ gives the
233 same object. Agreement concerns the shared domain; descriptions on distinct
234 parts can remain different. For the finite spatial measures used here, equal
235 overlap restrictions give a unique measure on the covered domain. This is
236 the gluing property formalized by a sheaf.

237 In the cup example, suppose a visual description locates the cup on a
238 table and a reaching description relates the hand to the cup. To compare
239 them, a model must express their shared quantity, such as cup location, in
240 a common coordinate system. Restriction would retain that quantity while
241 dropping details peculiar to each description. Agreement on it permits the
242 visual description to contribute surface appearance and the reaching descrip-
243 tion to contribute hand configuration without duplicating the whole situa-
244 tion. Disagreement after coordinate conversion would make the two estimates
245 incompatible at that level of description, however closely timed their activity.

246 The formal counterexample makes the additional requirement visible. Ev-
247 ery patch has the same oscillator phase, yet two local mass profiles assign
248 different densities to a shared site. They cannot be restrictions of a single

249 global measure. Phase order therefore cannot establish compatibility. Con-
 250 tinuity of a physical field does not repair disagreement in the descriptions
 251 associated with it.

252 Agreement to within a tolerance determines less than a glued state, and
 253 more than nothing. Suppose the local profiles differ by at most ε in the
 254 uniform distance at every shared site. Nonnegative weights that vanish off
 255 their own patch and sum to one at each site then select a global profile lying
 256 within ε of every patch on that patch, and any two such selections differ by
 257 at most ε , with no dependence on how many patches cover a site; at $\varepsilon = 0$
 258 the selection is the unique glued state. What the tolerance costs is existence:
 259 for every positive ε there are families at that tolerance admitting no global
 260 description at all, so selection returns a choice among states rather than an
 261 extension of the data. A readout that is L -Lipschitz in the state varies by at
 262 most $L\varepsilon$ across selections, which is the resolution condition an interpretation
 263 at scale δ would have to meet. The supplement states these theorems and a
 264 three-site witness that attains the bound. Selection supplies no process by
 265 which local descriptions come to agree, and no rule that picks the cover or
 266 the weights.

267 For a theory of experience, the attraction of gluing is this contribution
 268 of different local contents to one description. The cup example does not yet
 269 instantiate the theorem: receptive territories and content domains must be
 270 related to the spatial patches, and their descriptions must have the required
 271 extension property. In particular, a decoder assigning probability laws to per-

ceptual contents needs a further extension model, because matching overlap
 marginals need not determine a unique global joint distribution (Abramsky &
 Brandenburger, 2011). The spatial-measure construction provides a precise
 case of gluing; a content model must justify its own existence and uniqueness
 claims.

4.2 Coherent adaptation can lose environmental alignment

An adaptive coupling rule might appear to supply that missing representa-
 tional structure. The squared-drift control shows why the objective mat-
 ters. Symmetric coupling conserves mean drift: pairwise sine terms cancel
 when summed over nodes. Minimizing the sum of squared drifts at that
 fixed mean therefore rewards equalizing drift across nodes. In the chosen
 frequency-cluster model, this favors coupling between unlike-frequency clus-
 ters, whereas the diagnostic template rewards coupling within like-frequency
 clusters. The objective and the scored structure pull in different directions.

For fixed phases, the continuum objective is

$$\sigma(K) = \frac{1}{D} \int \left[\omega(x) + \int K(x, y) \sin(\theta_y - \theta_x) d\mu(y) \right]^2 d\mu(x). \quad (6)$$

The formalization constructs the bounded drift operator A and its adjoint,
 obtains $\nabla\sigma = (2/D)A^*(AK + \omega)$, and proves descent for the specified gradient
 flow. This is a squared-drift functional; identifying thermodynamic entropy

291 production would require probability-current information.

292 The finite control co-evolves noisy phases and symmetric, nonnegative
293 couplings at fixed total resource. The frequency-cluster template is a di-
294 agnostic covariance of shared latent signals, not a training input. Across
295 three seeds, second-half order remains between 0.9167 and 0.9222, while the
296 within-cluster to between-cluster coupling ratio falls from 0.9824–0.9948 to
297 0.3507–0.6135 (Fig. 4). The supplement reports random-update, relabelled-
298 gradient and frozen-kernel controls, including comparisons at overlapping
299 deformation and coherence ranges and against partitions blind to frequency.

300 The objective reduction is transient: averaging over complete update
301 intervals removes the apparent gain seen immediately after updates. A pre-
302 specified grid of 6 rate–cadence combinations contains no case meeting the
303 declared 5% sustained-reduction criterion while retaining coherence and los-
304 ing alignment. The finding is coherent adaptation away from the specified
305 structure, with no demonstrated sustained cost reduction.

306 Kernel alignment and overlap compatibility are different observables.
307 This control tests the narrower suggestion that coherent adaptation under
308 this objective acquires the designated environmental structure. It fails for
309 an intelligible reason: the objective rewards a different organization. A
310 thermodynamically specified learning model can include data in its objective
311 (Corberi et al., 2026); explaining agreement about the cup would likewise
312 require a learning rule connected to the relevant content.

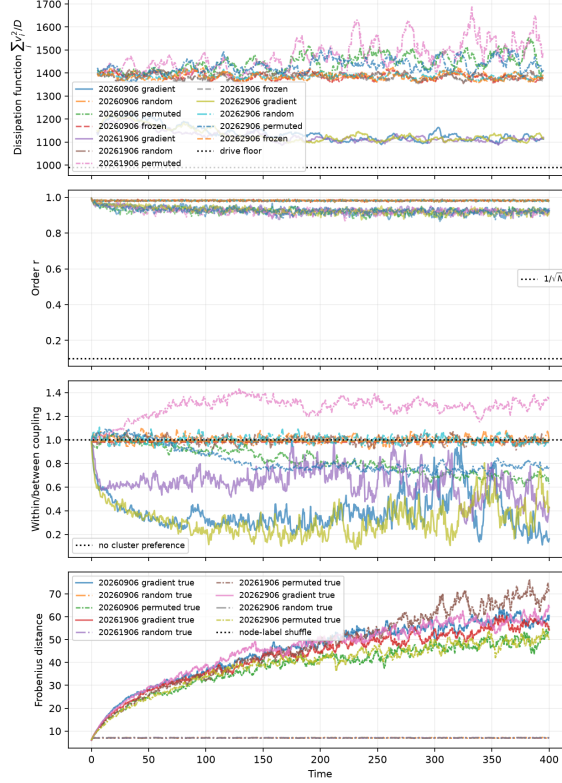


Figure 4: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), relabelled gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

313 4.3 What must be specified and measured

314 Compatibility enters the construction as a hypothesis about the local states.
315 For deterministic positive coupling, convergence to a potential minimum is
316 proved under an explicit initial basin condition; the proposed bridge E78
317 connects supercritical noisy coherence to relaxation of the specified cover.
318 Neither step explains how initially inconsistent representations become con-
319 sistent. That is the mechanism a biological account must add.

320 A neural test must first identify the domains and objects whose agreement
321 is at issue. Two decoders should estimate a shared content on overlapping
322 domains, with restriction maps specified before inspecting their agreement.
323 Their discrepancy needs a null under decoding noise, and the procedure must
324 distinguish deliberately constructed compatible and incompatible examples.
325 A common prior can make separate decoders agree artificially; region-specific
326 bias can make them disagree artificially. The supplement develops these re-
327 quirements. An estimator built from phase agreement would miss the prop-
328 erty the counterexample separates.

329 Choosing a cover is also part of choosing a candidate subject. Trivial
330 descriptions can agree while carrying little content, and a suitably chosen
331 partition can conceal disagreement. The scientifically useful choice must
332 therefore be constrained independently by what local activity encodes and
333 which shared quantities can be compared. The uniqueness theorem then
334 answers a precise question about that choice: whether the supplied local
335 descriptions determine one global state. It leaves the empirical selection of

domains and descriptions to the model.

5 Self-representation: a state that models it-self

5.1 The reflexive construction

Compatibility yields a global state. Reflexivity adds a relation between that state and an encoding within it. Equip global sections with a complete metric. Under the additional restriction-resonance condition, the encoding of a section s is its restriction to a region U , and the map takes the form

$$F(s) = \text{readout}(s|_U). \quad (7)$$

The same region and readout are used across the model's state space. If F contracts and fixes the section s_* supplied by the cover, Banach's theorem makes s_* its unique fixed point (Banach, 1922). The formal bridge E89 supplies the contraction law and $F(s_*) = s_*$ for the boundary's encoding-readout composite. Identifying that encoding with restriction is a separate resonance condition; E89 does not supply it. The supercritical order parameter provides an inequality used by the specified contraction bound; constructing the encoding and readout remains a separate obligation.

The equality ties the reconstruction to the state formed from the local

353 descriptions. In the cup example, the candidate global state would include
354 the represented scene and the perceiver’s relation to it. A local encoding
355 would have to retain enough information for the specified readout to recover
356 that modelled state. An encoding of cup position alone would be insufficient
357 if the readout could not recover the represented hand and viewpoint. The
358 term *predictive* names this reconstruction map; a temporal neural prediction
359 requires a dynamical interpretation.

360 **5.2 Why connect these properties to unity and a min-** 361 **imal self?**

362 We identify the section glued from compatible local states with the unity
363 of a conscious episode, and its accurate internal reconstruction with a mini-
364 mal self. Two features of the episode motivate this interpretation: different
365 contents belong to one situation, and that situation is experienced from a
366 perspective.

367 The unity proposal starts from a distinction between having several con-
368 tents and having several experiences. Seeing the cup’s surface and feeling
369 the reaching hand contribute different contents to one situation. Compat-
370 ibility expresses their agreement about what they share while preserving
371 those differences. An episode can also contain uncertainty or conflicting
372 judgments. A model might represent the cup’s apparent position separately
373 from a judged position and consistently describe the discrepancy. Deciding

374 whether two estimates concern the same quantity or distinct aspects of ex-
375 perience is therefore part of defining the content model. A choice made only
376 to ensure agreement would give the unity proposal no empirical force.

377 The motivation for a minimal self is that the cup is encountered from a
378 location and in relation to a body that can reach it. The proposed internal
379 representation concerns the perceiver’s condition in that encounter: where
380 the perceiver is situated, what is being represented and how the body relates
381 to it. Such a representation could be implicit in the organization of activity;
382 it need not describe a person verbally or carry an autobiographical narrative.
383 Requiring it to reconstruct the same global state links the perspective to the
384 episode whose contents it organizes.

385 The fixed-point condition requires accuracy about the very state contain-
386 ing the encoding. Reconstruction alone is insufficient for a perspective: a
387 recording can be reconstructed without being used by a system to represent
388 its own condition. The proposed interpretation therefore needs evidence that
389 the encoding helps the system distinguish changes in itself from changes in
390 its surroundings. Changing the encoded hand or viewpoint while holding
391 the external scene fixed should alter the system’s estimates of its relation
392 to the cup as the readout predicts. This functional criterion goes beyond
393 $F(s_*) = s_*$; whether the represented perspective is experienced remains a
394 further question.

395 Accuracy concerns the variables represented by s_* , rather than every mi-
396 croscopic physical fact. For a fixed map F with Lipschitz factor $q < 1$ and

397 fixed point s_* , approximate reconstruction has a quantitative bound:

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1 - q}. \quad (8)$$

398 At the stipulated rate $q = e^{-(K-2D)\tau/2}$, this radius grows as $2\varepsilon/((K-2D)\tau)$
399 near threshold, for fixed positive ε and τ . The bound weakens while the
400 exact fixed point remains unique. The supplement gives explicit inequalities
401 and composes reconstruction error with displacement of the state. A cortical
402 realization must specify the represented variables, their metric and the
403 required fidelity; changing readouts require a further dynamical model.

404 We propose reflexivity as necessary because even an unreflective episode
405 has a perspective: the cup appears somewhere relative to the perceiver,
406 whether or not the perceiver attends to themselves. Requiring internal self-
407 encoding on that basis is the philosophical hypothesis. Gluing alone does
408 not require it; the dissociations below test the two commitments separately.

409 5.3 Which cover and which self-model?

410 Both uniqueness results are conditional on their objects. Gluing gives one
411 global extension of specified compatible local data; Banach gives one fixed
412 point of a specified contracting map. These solve ambiguity within a chosen
413 construction. They do not choose between different covers, different encod-
414 ings or different candidate subjects. Integrated information theory makes
415 exclusion an explicit part of its account (Tononi et al., 2016); the present

416 uniqueness results do not settle that selection problem.

417 The relevant cover and map must be tied to neural organization by evi-
418 dence independent of the desired conclusion. Local descriptions should pre-
419 dict the contents attributed to their domains, and their restrictions should
420 preserve meaning on shared domains. The encoding and readout should be
421 specified before testing them, with their sensitivity to changes in the pro-
422 posed state measured over a declared range.

423 A constant readout returning one predetermined state contracts and has a
424 fixed point, even when it discards an encoding supplied by restriction. Such a
425 construction fails the sensitivity requirement above. This is why selecting the
426 map requires evidence about its functional role in addition to mathematical
427 existence.

428 **5.4 Dissociations that would bear on the identification**

429 A test of the full commitment must measure its clauses separately: effective
430 coupling relative to diffusion, overlap compatibility, and the accuracy of the
431 proposed self-encoding, alongside an independent endpoint for experience.

432 A case meeting all three physical and representational conditions with evi-
433 dence against a unified conscious episode would challenge their sufficiency.

434 Unified experience with demonstrably incompatible local descriptions would
435 challenge the unity identification. Unified experience with compatibility but
436 without the specified self-encoding would challenge the reflexive requirement.

437 Anesthetic emergence, dreaming and dissociative states are candidate set-

438 tings because the framework asks whether coordination, unified content and
 439 self-representation can separate. Interpreting such evidence requires a de-
 440 clared relation between the experience endpoint and report; absence of report
 441 alone cannot define absence of an episode. The compatibility observable in
 442 the supplement is a specification, and the corresponding neural self-encoding
 443 has yet to be identified. These are the main obstacles to testing the inter-
 444 pretive commitment.

445 The proposal concerns which state is conscious. It supplies no explanation
 446 of why satisfying these relations should be accompanied by experience. Nor
 447 do its gluing and fixed-point conditions select an electromagnetic substrate:
 448 the cortical field commitment resides in the physical identification and cali-
 449 bration. Their multiple realizability leaves the status of other biological and
 450 artificial implementations open.

451 **6 Thermodynamic constraints and the con-** 452 **ditional composition**

453 The broader composition constrains one proposed physical implementation
 454 through finite-register capacity $H(p) \leq \log |X|$, erasure heat under a specified
 455 system–bath model (Landauer, 1961; Norton, 2005; Shenker, 2000), and a
 456 separate field-boundary result for disconnected vacuum components (Kibble,
 457 1976). The supplement develops these results and their witnesses. The phase
 458 threshold and representational argument stand independently of a derivation

459 from finite capacity.

460 The constraint most directly relevant to representation concerns predic-
 461 tive memory. For a state X_t driven by an environmental signal S_t ,

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (9)$$

462 Under the no-back-action Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$, the bracket is
 463 nonnegative (Still et al., 2012). The bound limits nonpredictive memory at a
 464 given dissipation budget; zero retained information satisfies it too. Relating
 465 it to Landauer heat requires that memory waste be bounded by erased en-
 466 tropy at the same thermal scale and heat budget. These distinct information
 467 constraints (Rosas et al., 2026) leave content and learning dynamics to be
 468 specified.

469 An acting agent changes its future inputs, so the passive assumption
 470 requires justification for the variables and interval considered. A feedback
 471 model specifies a policy $A_t \sim \pi(\cdot | X_t)$, world dynamics $S_{t+1} \sim K(\cdot | S_t, A_t)$,
 472 an observation channel and a memory update. For finite states, the exact
 473 information balance is

$$I(X_t; S_t) - I(X_t; S_{t+1}) = I(X_t; S_t | S_{t+1}) - I(X_t; S_{t+1} | S_t). \quad (10)$$

474 Both conditional terms are nonnegative. The last term vanishes for the
 475 passive model; with feedback, the signed difference can be negative because

actions create correlations with the future. Its causal interpretation requires an environmental state sufficient to account for the relevant dynamics.

The finite thermodynamic extension instead bounds joint entropy reduction. For an elementary update that holds the controller fixed, strictly positive forward and reverse path probabilities give $\Sigma = D_{KL}(P_F||P_R) = \Delta H(X, S) + \langle Q \rangle / (k_B T) \geq 0$ under local detailed balance, with Q positive into the reservoir (Horowitz and Esposito, 2014; Ito and Sagawa, 2013). This supplies the thermodynamic constraint for the active branch below. A proved two-substep extension composes actuation with a memory update initialized from its coordinate-swapped final law. It identifies the output with the perception–action channels and cancels the same intermediate entropy and expected energy. At a common thermal scale, the whole update’s joint entropy decrease is bounded by its total heat budget divided by $k_B T$, without assuming stationarity. The supplement gives the balances and reciprocal two-bit witnesses with explicit interaction energy, heat and work. Action and observation alias the existing states; additional physical registers require further accounting.

An explicit objective also permits finite policy selection. For a supplied terminal reward and a finite set of deterministic policies, a nonempty set satisfying the complete update’s heat budget has a reward maximizer. A two-bit lamp task has a unique feasible optimum that strictly improves on a feasible constant-action baseline; a stronger policy exceeds the budget. Reward, heat and work are evaluated on each policy’s actual composed process, with

499 common initial law, channels, energy and reservoir convention. The base-
500 line meets the heat budget while missing the task target, so the objective
501 supplies content that thermodynamics alone does not. This result compares
502 fixed policies for one update.

503 A separate finite learning model specifies how a policy register adapts. Its
504 bit selects between the two feasible lamp policies, and its energy decreases
505 with their known expected reward. An autonomous thermal channel changes
506 the register, carrying each final distribution into the next update. From the
507 specified initial preparation, Lean proves strictly increasing expected task
508 reward and geometric convergence to a stationary mixture. Transition heat
509 obeys local detailed balance, and cumulative update heat equals the regis-
510 ter’s finite energy loss at zero external work. The mixture remains below the
511 optimal policy; individual updates can lower reward, and a different initial
512 preparation lowers expected reward. This is adaptation toward an encoded
513 objective. Preparing the register and implementing its readout in an actua-
514 tor require additional accounting. Task reward and task heat are prospective
515 expectations under the prepared lamp law; the learning clock executes no
516 task episodes and supplies no spatial coupling convergence or cortical iden-
517 tification.

518 A second learning model supplies no values at all. Which of two actions
519 is rewarded is an environmental bit that no channel of the learner takes as an
520 argument: the register’s update is a function of the observed outcome alone,
521 and its readout, which fixes the action executed, is a function of the register

522 alone. The world returns success with probability $3/4$ for the rewarding
 523 action and $1/4$ for the other; success leaves the register where it is and failure
 524 redraws it. Acting, observing and updating compose into one channel on the
 525 joint law of parameter and register, so reward, heat and entropy are read from
 526 the same process. From a uniform prior over both bits, Lean gives every mass
 527 of that law at every horizon: each agreeing pair carries $3/8 - \frac{1}{8}(1/2)^n$, so the
 528 expected reward of the action actually executed is $3/4 - \frac{1}{4}(1/2)^n$, strictly
 529 increasing from $1/2$ and always short of certainty. A frozen register and an
 530 uninformative world, in which success has probability $1/2$ whatever is done,
 531 both hold $1/2$ at zero cost, so the improvement is bought by the observations
 532 rather than by the update rule. Cumulative heat is $(\log 3/4)[1 - (1/2)^N]$, and
 533 the register's two states are energetically degenerate, so that heat is external
 534 work. It is irreducibly so: a register energy that made the updates free would
 535 force the drive's heat to be equal at both parameter values, and this drive's is
 536 not. The learner acquires no preference between the actions, each of which it
 537 executes half the time at every horizon, but a correlation with the parameter.
 538 Preparing the prior, supplying the work, executing further task episodes and
 539 fabricating the readout are separate accounting, and no cortical identification
 540 follows.

541 **6.1 What the composition establishes**

542 A continuing finite agent sequences actuation, observation-driven learning
 543 and reset on one joint law of an unknown task parameter, a register and a

544 task flag. The register drives the actuator; the observation reads the flag
 545 that action changed; the reset prepares its next use. Each stage starts from
 546 the preceding output. For a supplied initial work store B , a common energy
 547 E and thermal scale θ , the first law and entropy balance telescope to

$$\theta[H(p_0) - H(p_N)] \leq Q_{<N} = W_{<N} + \langle E \rangle_0 - \langle E \rangle_N \leq B + \langle E \rangle_0 - \langle E \rangle_N, \quad (11)$$

548 provided expected cumulative work stays within B at every prefix and each
 549 stage satisfies positive support and local detailed balance. A bit witness
 550 retains agreement with the unknown rewarding action above chance after
 551 every completed cycle, while every cycle draws a positive minimum of work,
 552 including reset. Blinding the observations leaves chance performance and
 553 makes the learning stage free. Replacing reset by idle drift preserves some
 554 learning but reduces the next cycle's performance. A finite store therefore
 555 supports only a bounded prefix of this protocol. This is accounting in expec-
 556 tation: the initial law and store are supplied resources, and the observation
 557 is marginalized inside the learning channel. Pathwise stores, a separate sen-
 558 sor memory and preparation channels are treated below. Heat is assigned to
 559 the composite learning channel through local detailed balance, and the store
 560 bounds a mean; no cortical identification follows.

561 A store the protocol carries rather than reports supports a different claim.
 562 Let the state include the store's reading β , and let each executed transition

563 draw w^- from it and receive a declared supply w^+ , with

$$\beta(t) = \beta(s) - w^-(s, t) + w^+(s, t), \quad w^-(s, t) \leq \beta(s) + w^+(s, t) \quad (12)$$

564 on every transition of positive probability out of a state the protocol reaches.
 565 A reading that starts nonnegative and satisfies the second condition at every
 566 stage is nonnegative at every state the protocol reaches, at every horizon.
 567 Averaging the first identity over the executed paths returns the mean ledger,
 568 $\langle \beta \rangle_N = \langle \beta \rangle_0 - W_{<N}^- + W_{<N}^+$, so pathwise solvency implies the expected bound
 569 Eq. (11) assumes. The converse fails: a store that spends a unit or idles with
 570 equal probability covers its expected draw at both prefixes of a two-stage
 571 run, with equality at the second, and is overdrawn on the path of probability
 572 $1/4$ that spends twice. An expected allowance is a bound on a mean and not
 573 a battery. Two conditions fence sustained operation. If the supply covers
 574 each draw as it is made, every horizon is solvent and no bound on cumulative
 575 work is needed. If instead $w^- - w^+ \geq c > 0$ on every reachable transition,
 576 a reading starting no higher than β_0 gives $Nc \leq \beta_0$, which is the pathwise
 577 form of the store bound and constrains each trajectory rather than a mean.
 578 A deterministic spend-and-recharge cycle satisfies neither condition and is
 579 solvent at every horizon while the work it has drawn exceeds any declared
 580 allowance, so continuing operation is a claim about the supply. None of this
 581 requires positive support; the generic store accepts its supply as an input.

582 A finite source can be included in the same state. If its usable energy R

583 pays for each supplied transfer, then on every reachable transition

$$R' = R - w^+, \quad (\beta + R)' = (\beta + R) - w^-. \quad (13)$$

584 Internal replenishment cancels from the combined ledger. Its expected supply
585 is $W_{<N}^+ = \langle R \rangle_0 - \langle R \rangle_N$, and nonnegative store and source readings give
586 $W_{<N}^- \leq \langle \beta + R \rangle_0$. If every executed draw before horizon N is at least $c >$
587 0 and the initial combined reading is at most b , then $Nc \leq b$. A direct
588 transfer of one unit followed by idling saturates this bound; the cost premise
589 concerns the finite run. Available energy does not guarantee delivery: a
590 finite witness starts a source bit charged and its buffer and output load
591 empty, with a uniform selector bit. The selector controls a source–buffer
592 swap, followed by a buffer–load swap. Both gates are reversible and conserve
593 the three energy bits’ total energy. Half the paths deliver a unit to the load;
594 half leave that energy in the source. The actual mean draw is $1/2$, and
595 claiming a unit on every path violates the store ledger. An initially empty
596 source delivers nothing. The indefinitely replenished charger cannot have
597 any nonnegative finite source coordinate satisfying Eq. (13). This witness
598 models one uncertain transfer followed by idling; source preparation, gate-
599 control costs, external refuelling and identification with the learning agent’s
600 actuator remain separate.

601 The measurement has a register of its own, and clearing it is an executed
602 operation. Let the state carry a memory coordinate alongside the register and

603 the world, and run four operations in sequence: the readout drives the actua-
 604 tor, a measurement writes the memory from the world the action moved, the
 605 register is updated from the memory alone, and a clearing operation drives
 606 the memory toward a declared standard law ν while the world is prepared.
 607 The update takes the memory and not the world, so two worlds holding the
 608 same memory drive the register identically: what the register can learn is
 609 what the memory carries. The clearing operation drives three coordinates at
 610 once, and its transition log-ratio is the sum of the three coordinates' own, so
 611 the memory's share is a named part of that operation's cost rather than a
 612 residue. For an erasure that lands on ν whatever it is given, that share is

$$\langle Q_M \rangle = \theta[H(\mu) - H(\nu)] + \theta D_{KL}(\mu||\nu), \quad (14)$$

613 with μ the memory's marginal before the operation. The first term is Lan-
 614 dauer's bound and the second is the cost of applying one fixed channel to a
 615 law it is not matched to; Gibbs' inequality turns the identity into the bound.
 616 The identity also states what the bound does not. A memory already at ν
 617 is free to clear, and one more ordered than ν draws heat out of the reser-
 618 voir, so clearing dissipates because of what the memory holds rather than by
 619 construction.

620 A bit witness fixes the numbers without computing the joint law. The
 621 memory's marginal after a measurement is a mixture of measurement outputs
 622 over a world marginal that is itself a mixture of actuator outputs; with every

623 channel mass $1/4$ or $3/4$, the memory's mass on a set flag lies in $[7/16, 9/16]$ at
 624 every clearing, whatever the agent has learned. At $\nu = (3/4, 1/4)$ the share is
 625 $[3/4 - \mu(0)] \log 3$, hence between $(3/16) \log 3$ and $(5/16) \log 3$ at every cycle:
 626 no amount of learning makes the measurement free. In the first cycle the
 627 memory is exactly uniform and the two terms of Eq. (14) are $(3/4) \log 3 - \log 2$
 628 and $(1/2) \log(4/3)$, summing to $(1/4) \log 3$; the first is positive because $16 <$
 629 27 . Two controls fence the reading. A blind measurement that records
 630 nothing about the world leaves the same marginal and pays the identical heat,
 631 so the cost is set by the memory's marginal and not by what the memory
 632 is about. Replacing the clearing by the symmetric drift costs nothing and
 633 carries the measurement into the next episode, where the erasure returns the
 634 standard state whatever was written. The four operations are stages of one
 635 protocol, so Eq. (11) charges the whole run, clearing included. Fabricating
 636 the gates and any cortical identification remain outside this model; preparing
 637 the prior is charged below by the same identity.

638 The work those operations draw can be carried in the same state. Let a
 639 coordinate R of the agent's own state read a finite charge, let each executed
 640 transition draw that coordinate's loss, and let the clearing spend at least c of
 641 it while no operation returns any. If the charge starts no higher than b and
 642 stays nonnegative through m complete cycles, then

$$mc \leq b, \tag{15}$$

643 so a finite source funds finitely many clearings. The bound needs a cost that
 644 depends on the stage: an agent that pays at one operation in four meets
 645 a uniform per-transition cost only at $c \leq 0$, while a charge paid once in
 646 each period of p operations sums to mc over m periods however it sits inside
 647 the period. A witness whose world carries the task flag and a charge of
 648 three units, one spent at each clearing, attains Eq. (15) at three cycles and
 649 reads below empty at the sixteenth operation. Its clearings cost between
 650 $(3/16)\log 3$ and $(5/16)\log 3$ throughout, by the marginal bounds above, so
 651 what the source funds is what Eq. (14) prices. The drawn unit is a declared
 652 reading of the state and is not identified with that heat.

653 Two limits here belong to the model rather than to the witness. The
 654 clearing is a channel of the memory alone, so it cannot be conditioned on
 655 the charge: past the third cycle the agent still clears, still costs at least
 656 $(3/16)\log 3$ and draws nothing, so the run is unfunded rather than halted.
 657 And an agent all of whose channels have full support cannot be funded at
 658 all. A ledgered reading that never rises is then constant between every pair
 659 of states, in both directions, so every net draw is zero, and the witness's
 660 restricted support is forced by that rather than chosen. Preparing the charge,
 661 refuelling it from outside and fabricating the gates remain outside this model.

662 Preparation can be modelled by the same constant channel as memory
 663 clearing. For a strictly positive target ν , Eq. (14) gives its mean log-ratio
 664 heat from hardware law μ as $\theta[H(\mu) - H(\nu)] + \theta D_{KL}(\mu||\nu)$. Against a uni-
 665 form target the two terms cancel, because $D_{KL}(\mu||\text{unif}) = \log |S| - H(\mu)$:

666 the channel has zero mean heat from any starting law. The reservoir inter-
 667 pretation uses local detailed balance; gate construction, control work and a
 668 funding source require separate accounting. Prepending this channel as stage
 669 zero includes preparation in the cumulative balance whenever its hypotheses
 670 hold. Once it reaches the declared prior, every subsequent law is the original
 671 agent's own. The sensor-memory witness prepares its uniform prior from a
 672 different, strictly positive hardware law and satisfies the whole-run entropy
 673 bound.

674 Two things a preparation cannot do are properties of its signature. It
 675 writes the system coordinate and leaves the parameter's marginal as it found
 676 it, so the agent's uncertainty about the world is an input rather than some-
 677 thing it prepares; and a preparation that reads neither coordinate produces
 678 a product, so it cannot create a correlation between parameter and regis-
 679 ter. At $\theta = 1$, preparing a bit law biased to ε from uniform has mean heat
 680 $(1/2 - \varepsilon) \log[(1 - \varepsilon)/\varepsilon]$ for $0 < \varepsilon < 1$; this family has no finite upper bound. Its
 681 sharp endpoint has zero reverse probabilities and fails the identity's positive-
 682 target premise. This is a restriction of the specified channel and heat model;
 683 it supplies no cost bound for other implementations of exact preparation,
 684 including the charge above.

685 Table 1 lists the connecting hypotheses of two branches. The theorem
 686 **chain** uses passive predictive memory; **chain_active** uses a specified finite
 687 feedback process. Each accepts eight implications and shares the passage
 688 from coarse-graining to the phase model and its proposed representational

689 role. In both, the finite register and the field boundary are independent
690 starting objects.

691 For the active branch, the controller has the register's state type and the
692 fourth node records

$$\theta > 0, \quad \theta[H(p) - H(p')] \leq \langle Q \rangle \leq B, \quad (16)$$

693 where p, p' and $\langle Q \rangle$ belong to that same controlled substep. E34Active sup-
694 plies positive path probabilities, local detailed balance at the register's ther-
695 mal scale θ , and the allocation $\langle Q \rangle \leq B$, with B equal to the register's
696 heat. Landauer's lower bound does not imply this upper-budget allocation
697 or identify erasure with actuation. A resource model of one register supplies
698 it instead: every operation is controlled by that register and exchanges heat
699 with its reservoir at θ , and the register's dissipated heat is the total its op-
700 erations deliver there. An operation that does not increase the joint entropy
701 then has a nonnegative share, by the left inequality of Eq. (16) applied to it,
702 so an operation whose companions all compress has a share no larger than
703 the register's dissipation. That accounting identity and that compressiveness
704 are the model's physical inputs; the allocation follows from them. Composing
705 the operations into one evolving law with its preparation and power source
706 is a separate model, and the register-bath model at the end of this sub-
707 section shows that a microscopic bipartite dynamics supplies neither input.
708 E45Active requires convergence of the specified coupling energies. One scalar

709 constitutive model connects those energies to the named feedback step: writ-
 710 ing $d = H(p) - H(p')$, it sets the spatial density to $f_A(x) = g d b(x)$, with
 711 supplied gain $g \geq 0$ and profile b . Regular spatial refinement gives

$$E_n(f_A) \longrightarrow L_A = g d \int_S b d\mu, \quad L_A \leq g \frac{B}{\theta} \int_S b d\mu \quad (17)$$

712 when $\int_S b d\mu \geq 0$. The limit follows from mesh assumptions; the upper
 713 bound uses Eq. (16) for the same process. The constitutive law uses ensemble
 714 entropy reduction and supplies neither a microscopic actuator nor its work
 715 account. Changing the gain or profile at fixed process and heat changes the
 716 coupling energies.

717 A second, microscopic model supplies an actuator and a kernel. The
 718 hardware carries finitely many spatial response modes with fixed profiles ϕ_i
 719 and installation prices u_i . A configuration z of the controller and its switches
 720 sets the mode occupancies $c_i(z)$, and those same occupancies determine both
 721 a reciprocal kernel on the product space and a stored installation energy,

$$K_z(x, y) = \sum_i c_i(z) \phi_i(x) \phi_i(y), \quad U(z) = \sum_i u_i c_i(z). \quad (18)$$

722 A transition $z \rightarrow z'$ changes K only through the modes whose occupancy it
 723 changes, so a pair outside the response of those modes is untouched, and the
 724 transition's external work is $U(z') - U(z) + Q$. The kernel the executed step
 725 installs is the average of K_z over the law that step actually leaves; its change is
 726 the mean of the pathwise changes, and the mean of the transition works splits

727 into the change in stored energy and the channel's heat. A finite measurable
 728 partition of the substrate with one sample per cell reconstructs the installed
 729 kernel throughout each cell pair, and the cell-mass weighted finite energy
 730 equals the product-measure integral of that reconstruction exactly. On a
 731 compact substrate with continuous profiles, a sample error tending to zero
 732 gives

$$\mathcal{E}_n \longrightarrow \iint K d\mu d\mu. \quad (19)$$

733 The refinement index here is spatial and is independent of the number of ex-
 734 ecuted control updates; convergence in one is not convergence in the other.
 735 The same hardware discharges the passive edge from a declared predictive
 736 system's own joint law, and two declared systems with different dissipated
 737 work give the identical edge, so predictive efficiency selects neither the hard-
 738 ware nor the partition. Mode profiles, prices, occupancy readout, reservoir
 739 convention and substrate are declared hardware data, which no heat bound
 740 supplies. Fabricating that hardware, powering it beyond one prepared up-
 741 date, and identifying the substrate or the modes with a physical field remain
 742 separate requirements.

743 A third finite model examines the register's own bath. One register bit
 744 and one bath bit carry each of the four deterministic maps of the register as
 745 a permutation of the pair, acting on a bath prepared in its ground state: an
 746 injective map moves the register alone, and an erasure swaps the register's
 747 bit into the bath. With bath levels 0 and $2 \log 2$, degenerate register levels, a

748 uniform register and $\theta = 1$, the bath's mean energy gain along the executed
749 paths is 0 for each injective map and $\log 2$ for each erasure, and those values
750 inhabit Landauer's heat equation with the bath supports the gates actually
751 reach. For an arbitrary finite protocol and an explicitly modelled bath energy
752 E_B ,

$$\sum_{k < N} \langle \Delta E_B \rangle_k = \langle E_B \rangle_N - \langle E_B \rangle_0, \quad \sum_{k < N} \langle q_k \rangle = \langle E_B \rangle_N - \langle E_B \rangle_0 + \sum_{k < N} \langle q_k - \Delta E_B \rangle, \quad (20)$$

753 where the first identity needs no positive support and supplies no entropy
754 inequality, and the second is the correction a reported heat ledger carries
755 when it omits part of that transfer. Three consequences fence what a map-
756 indexed budget can mean. Executing the same erasing gate twice returns
757 the $\log 2$ and restores the initial law, so the two shares are $\log 2$ and $-\log 2$
758 about a total of zero, and a reusable eraser requires its bath prepared again. A
759 second reversible gate holds the register fixed and excites its bath, performing
760 the same register map as idling at $2 \log 2$ instead of 0; the work supplied is
761 $\log 2$ for one erasure, $2 \log 2$ for that drive and zero for two erasures. And a
762 swap is an involution, so the gates' own transition log-ratios vanish on every
763 realized path while their bath gains $\log 2$: a deterministic reversible gate is
764 not a strictly positive reduced channel, and local detailed balance at any
765 thermal scale assigns it none of the heat its bath receives. Positivity, local
766 detailed balance and the ledger identity are inputs of the resource model
767 rather than consequences of microscopic reversibility. Preparing the bath,

supplying the work, and identifying this bath with the reservoir of the ledger's operations remain separate; the numerical values are those of the declared energy gap and preparation.

Two structures also enter both theorems: a thermodynamic cover, including overlap compatibility, and a reflexive boundary specifying the local encoding and readout. Given these and the connecting hypotheses, both conclude **UnifiedSelf**: the section glued on that cover is the unique fixed point of self-prediction. Restriction resonance, needed for the interpretation in Eq. (7), is a separate condition. The result binds the fixed point to the specified local states; compatibility and accuracy at that state are premises, while uniqueness follows from the construction.

The joint witnesses combine a double well, a register, a changing uniform-grid mesh and a three-site cortex; the active branch includes the noisy two-bit actuator. They establish that the hypotheses can hold together in distinct toy components. Mesh convergence is supplied independently of the thermodynamic bounds, and no common biological mechanism is established. A second active witness composes the same branch with the bridge discharged from that register's ledger: its two operations deliver $(2/5) \log 2$ and $(3/5) \log 2$ to one reservoir, summing to the register's $\log 2$. A third active witness closes one spatial response mode with a thermal switch: executing the channel strictly raises a pairwise kernel value, closing the mode costs its stored energy plus the channel's heat, and uniform partitions of an atomless interval give a coupling energy that changes between the first two refinements and

Connection	Status	Required assumption
E12	Independent physical premise	A finite-capacity system realizes a field that leaves its vacuum manifold. Capacity and the boundary are independent roots.
E23	Bridge assumption	The boundary is associated with a finite register whose update leaves states unreachable.
E34 / E34Active	Bridge assumption	Passive: a predictive structure at the register’s thermal scale and heat budget, with nonzero wasted memory. Active: positivity, local detailed balance and a register-heat allocation, derived from its ledger identity and other operations’ compressiveness. Preparation, source and store accounts use their own laws; identifying them with register heat remains an input. A reversible register–bath gate realizes erasure without satisfying the positivity or local-detailed-balance premise.
E45 / E45Active	Modelling assumption	Passive: a convergent regime, for hardware holding a declared predictive law. Active: Eq. (17) from a supplied gain, profile and meshes; Eq. (19) from declared response modes.
E56	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67	Physical commitment	That field operates above the synchronization threshold.
E78	Bridge assumption	Coherence drives a supplied cover to its relaxed equilibrium at coupling at least the named mean-field value.
E89	Modelling assumption	A map mediated by local self-encoding fixes the section glued on that cover and has the Lipschitz factor $q < 1$ used in the exact and approximate-state results.

Table 1: Connecting hypotheses of the passive and active compositions, eight per branch. They share E12, E23 and E56–E89. The cover’s compatibility is supplied as a structure field. Identifying the boundary’s encoding with restriction requires a separate resonance condition. The mathematical objects and node-level results are mapped in Supplemental Material, Table S1.

791 converges to the value the field witness uses. The active witnesses reject
792 a zero heat budget, an incompatible energy limit, a compressive operation
793 above the register’s whole dissipation, a ledger whose other operation draws
794 heat out of the reservoir, an uncharged installation, and a wrong spatial limit
795 for the microscopic sequence. Counterexamples also expose the role of over-
796 lap agreement. The development declares no additional Lean axioms beyond
797 the foundational axioms used by its libraries; physical premises are explicit
798 hypotheses and model fields.

799 **7 Discussion**

800 The strongest results concern what a coherence measurement leaves unre-
801 solved. Different recovery mechanisms share stationary distributions; equal
802 phases permit incompatible local descriptions; and the tested plasticity ob-
803 jective sustains coherence while moving away from its diagnostic structure.
804 These findings make the coupling hypothesis more demanding to test and
805 prevent a successful synchronization model from doing unexplained repre-
806 sentational work.

807 The proposed account identifies a compatible, internally represented state
808 with a conscious episode. Its main explanatory gap is the mechanism that
809 would maintain those relations: compatibility and accuracy at the glued state
810 enter the construction as premises. Their mathematical consequences leave
811 open why cortical activity should acquire or preserve them.

812 The most informative next step for that connection is a test of whether
813 coordination helps maintain independently measured content agreement. Af-
814 ter validating separate decoders and their overlap restrictions, a controlled
815 change in coupling or coordination could test whether agreement deterio-
816 rates in the predicted way, with controls for changes in sensory information
817 and decoder quality. Such a result would bear directly on the proposed role
818 of coherence. In parallel, the awakening hypothesis needs its own calibrated
819 geometry–coupling relation and resolved phase dynamics to distinguish a cou-
820 pling increase from diffusion change or common drive. A favorable recovery
821 trace alone would leave the representational connection open.

822 Testing the self clause requires perturbing a candidate encoding and com-
823 paring the resulting changes in represented scene, body and viewpoint with
824 the readout’s predictions. Establishing that functional role would make dis-
825 sociations from unified experience informative. The exact fixed-state ideal-
826 ization must also be assessed against the fidelity and time dependence of the
827 biological process.

828 Five components of the account are not derived here, and the formaliza-
829 tion marks each. Coarse-graining the discrete couplings converges a scalar
830 energy and produces no kernel: the vertex-sited candidate contributes ex-
831 actly zero continuum coupling energy on an atomless substrate, which is a
832 no-go for that architecture rather than for kernels, since cell pairs of pos-
833 itive mass do pass to a kernel limit. The field conversion $K_{\text{eff}} = \gamma N s E f$
834 carries its physical content in γ , a hypothesis about how long field-induced

835 timing shifts persist; the oscillation frequency, a phase relaxation rate and
836 the diffusion constant are candidates that need not agree, and the remaining
837 factors cannot choose between them. Descent of the squared-drift objec-
838 tive is proved for the specified flow at fixed phases, whereas the claim that
839 such adaptation acquires a designated environmental structure is not proved
840 and fails in the bounded study reported above. The non-identifiability re-
841 sult concerns stationary distributions, so it limits what a coherence trace
842 establishes rather than what any measurement can establish; separating the
843 mechanisms requires resolving dynamics. The thermodynamic results take a
844 prepared initial law as given and leave the supply of free energy to a sepa-
845 rate model. These are boundaries of scope and derivation. Reading any of
846 them as a positive finding about cortex, or as support for scale separation,
847 maladaptive learning or a limit on observation in general, would overstate
848 what has been checked.

849 Two of these results bear on formulations that read a generative model
850 off a system’s steady state, as the free-energy principle does (Friston, 2010).
851 Nothing here refutes that principle on its own terms: the squared-drift func-
852 tional is not a variational free energy, it is not identified with entropy pro-
853 duction, and no bridge between them is claimed or needed for any theorem
854 above. What transfers is narrower. The stationary distributions of increas-
855 ing coupling, decreasing diffusion and a matched common drive coincide, so a
856 steady-state density does not by itself determine which mechanism produced
857 it, and here the competing mechanisms are written down rather than sup-

posed. Separately, local descriptions agreeing pairwise on their overlaps need not determine one global state, and at every positive tolerance a three-patch family admits no global section at all; a decomposition into local descriptions therefore does not by itself deliver one belief about one world. Both points instantiate identifiability concerns already raised about that principle (Aguilera et al., 2022; Biehl et al., 2021; Bruineberg et al., 2022), in a model where the alternatives are explicit. Neither establishes that cortex fails to perform inference, and the plasticity control above bears on the objective it names and on no other.

Integrated-information, global-workspace and predictive-processing accounts organize unity, access and modelling differently (Tononi et al., 2016; Baars, 1997; Dehaene et al., 2014; Friston, 2010). The present proposal contributes a candidate interaction mechanism and explicit relations to test between local content and internal self-encoding. Its explanatory value will turn on whether those relations can be identified independently and shown to matter for experience. For the cup example, that means explaining how the system maintains one represented situation and how the perceiver’s place within it contributes to the episode.

Data and code availability

The Lean development, simulation scripts, tests and saved numerical summaries are maintained in the manuscript source repository. Publication sim-

879 ulation values are regenerated from saved summaries without rerunning pro-
880 duction sweeps. The exploratory EEG analysis uses OpenNeuro dataset
881 ds005620 (Bajwa et al., 2025); preprocessing and selection are described
882 in Supplemental Material. The source repository contains the reproducible
883 analysis and does not redistribute the raw recordings.

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887 **Use of AI tools**

888 AI assistance to this work was substantial and is not separable by task.
889 Several coding assistants and chat interfaces were used in an interleaved way
890 throughout: Google’s Antigravity CLI (Gemini 3.1, Claude 4.6 Opus, Claude
891 4.6 Sonnet), OpenAI’s Codex CLI (Astra, Terra, Luna, Sol), Claude Code
892 (Claude Opus 5), Gemini Pro through its web interface, and Fable (Claude
893 Opus 4) through OpenRouter, accessed with the Hermes CLI. Between them
894 they contributed to the Lean development, the simulation and analysis code,
895 and the drafting and revision of both documents. Because the tools were
896 used in combination and in alternation, no section, proof, figure or result is
897 attributable to any one of them, and none is claimed to be. The author is

898 responsible for all content, including everything a tool produced. Machine
899 checking establishes the stated Lean results under their hypotheses; it does
900 not validate biological interpretations or replace scientific review of generated
901 code and prose.

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